WORKED SOLUTION

STEP III 2011, Question 7

Pure • Proof by Induction

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q7

The key idea

Squaring $\sqrt{a+1}+\sqrt{a}$ gets rid of the separate square roots and leaves a number of the shape $p+q\sqrt{a(a+1)}$, so for even n we can step from T_n to T_{n+2} by multiplying by T_2. The quantity $A^2-a(a+1)B^2$ then turns out to be unchanged by that multiplication, which is exactly what the induction needs. One more factor of $\sqrt{a+1}+\sqrt{a}$ handles odd n, and the identities in (i) and (ii) say that the two terms are square roots of consecutive integers.

Solution

Throughout, write $N=a(a+1)$, so that $\sqrt{a+1}\,\sqrt{a}=\sqrt{N}$.

Part (i)

Base case, $n=2$. Expanding,

$$T_2=(a+1)+2\sqrt{a+1}\,\sqrt{a}+a=(2a+1)+2\sqrt{N},$$

so $A_2=2a+1$ and $B_2=2$ are integers, and

$$A_2^2-NB_2^2=4a^2+4a+1-4a(a+1)=1.$$

Inductive step. Suppose that for some even $n\geq 2$ we have $T_n=A_n+B_n\sqrt{N}$ with integers A_n, B_n satisfying $A_n^2-NB_n^2=1$. Write $p=2a+1$ for short, and note that $p^2=4a^2+4a+1=4N+1$. Since $T_{n+2}=T_nT_2$,

$$\begin{aligned}T_{n+2}&=(A_n+B_n\sqrt{N})(p+2\sqrt{N})\\&=(pA_n+2NB_n)+(2A_n+pB_n)\sqrt{N}.\end{aligned}$$

So we may take

$$A_{n+2}=pA_n+2NB_n,\qquad B_{n+2}=2A_n+pB_n,$$

which are integers because p, N, A_n and B_n are. For the identity, the cross terms $\pm 4pNA_nB_n$ cancel:

$$\begin{aligned} A_{n+2}^2 - NB_{n+2}^2 &= (p^2A_n^2 + 4pNA_nB_n + 4N^2B_n^2) - N(4A_n^2 + 4pA_nB_n + p^2B_n^2) \\ &= (p^2 - 4N)A_n^2 - N(p^2 - 4N)B_n^2 \\ &= A_n^2 - NB_n^2 = 1, \end{aligned}$$

using $p^2 - 4N = 1$ and then the inductive hypothesis. So the statement holds for $n+2$, and by induction it holds for every even positive integer n.

Note also that $A_2, B_2 > 0$ and the recurrences have positive coefficients, so $A_n > 0$ and $B_n > 0$ for every even n. We shall need this in part (iii).

Ask yourself Why does $A^2 - NB^2$ survive multiplication by T_2?

Because $A^2 - NB^2 = (A + B\sqrt{N})(A - B\sqrt{N})$, and the same calculation shows that the "conjugate" $A - B\sqrt{N}$ is multiplied by $p - 2\sqrt{N}$ at each step. So the product gets multiplied by $(p + 2\sqrt{N})(p - 2\sqrt{N}) = p^2 - 4N = 1$. The algebra above is just this fact written out.

Part (ii)

The case $n = 1$. $T_1 = 1 \cdot \sqrt{a+1} + 1 \cdot \sqrt{a}$, so $C_1 = D_1 = 1$, and $(a+1)C_1^2 = a + 1 = aD_1^2 + 1$.

The case $n \geq 3$ odd. Then $m = n - 1$ is even and positive, so by part (i) $T_m = A_m + B_m\sqrt{N}$ with $A_m^2 - NB_m^2 = 1$. Using $\sqrt{N}\sqrt{a+1} = (a+1)\sqrt{a}$ and $\sqrt{N}\sqrt{a} = a\sqrt{a+1}$,

$$\begin{aligned} T_n = (\sqrt{a+1} + \sqrt{a})(A_m + B_m\sqrt{N}) &= A_m\sqrt{a+1} + A_m\sqrt{a} + (a+1)B_m\sqrt{a} + aB_m\sqrt{a+1} \\ &= (A_m + aB_m)\sqrt{a+1} + (A_m + (a+1)B_m)\sqrt{a}. \end{aligned}$$

So $C_n = A_m + aB_m$ and $D_n = A_m + (a+1)B_m$ are integers (and positive). Now

$$\begin{aligned} (a+1)C_n^2 - aD_n^2 &= (a+1)(A_m^2 + 2aA_mB_m + a^2B_m^2) \\ &\quad - a(A_m^2 + 2(a+1)A_mB_m + (a+1)^2B_m^2) \\ &= A_m^2 + a(a+1)[a - (a+1)]B_m^2 \\ &= A_m^2 - NB_m^2 = 1, \end{aligned}$$

since the A_mB_m terms cancel. Hence $(a+1)C_n^2 = aD_n^2 + 1$, as required. (The case $n = 1$ is really the same calculation with $T_0 = 1$, that is $A_0 = 1$, $B_0 = 0$.)

Part (iii)

n even. Since $A_n > 0$ and $B_n > 0$,

$$T_n = A_n + B_n\sqrt{N} = \sqrt{A_n^2} + \sqrt{NB_n^2},$$

and by part (i) the integers NB_n^2 and $A_n^2 = NB_n^2 + 1$ are consecutive.

Ask yourself Why do we need the signs of A_n and B_n?

Because $\sqrt{A^2} = |A|$, not A. Writing $A_n = \sqrt{A_n^2}$ is only valid once we know $A_n \geq 0$, which is why we recorded positivity in part (i).

n odd. Since $C_n, D_n > 0$,

$$T_n = C_n\sqrt{a+1} + D_n\sqrt{a} = \sqrt{(a+1)C_n^2} + \sqrt{aD_n^2},$$

and by part (ii) the integers aD_n^2 and $(a+1)C_n^2 = aD_n^2 + 1$ are consecutive.

So for every positive integer n, $T_n = \sqrt{k+1} + \sqrt{k}$ for some positive integer k: namely $k = a(a+1)B_n^2$ for even n and $k = aD_n^2$ for odd n.

Checks

Take $a = 3$, so $T_n = (2+\sqrt{3})^n$. Then $T_2 = 7 + 4\sqrt{3} = 7 + 2\sqrt{12} = \sqrt{49} + \sqrt{48}$, and $49 = 12 \cdot 4 + 1$. Next, $A_2 = 7$, $B_2 = 2$ give $C_3 = 13$, $D_3 = 15$, and $T_3 = 26 + 15\sqrt{3} = \sqrt{676} + \sqrt{675} \approx 51.9808$, which agrees with $(2+\sqrt{3})^3$ computed directly. A computer ran the recurrences for $1 \leq a \leq 39$ and $n \leq 23$, confirming both identities exactly in integer arithmetic and the expansions of T_n numerically.

Common slips

- Forgetting the base case, or starting the induction at $n = 1$ when part (i) is only about even n.
- Writing $A_n = \sqrt{A_n^2}$ in part (iii) without noting that A_n (and B_n, C_n, D_n) are positive.

Beyond STEP

The equation $x^2 - Ny^2 = 1$ is Pell's equation, and part (i) shows that powers of $(2a+1) + 2\sqrt{a(a+1)}$ generate infinitely many solutions for $N = a(a+1)$. The key fact is that the "norm" $x^2 - Ny^2$ of $x + y\sqrt{N}$ is multiplicative, which is the start of the theory of quadratic number fields and their units. Davenport's *The Higher Arithmetic* has a very readable account of Pell's equation and continued fractions.

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